

QKS-v2 vs ViT4Mal [2] — Performance Comparison on IMG_DS (3,444 images, PYNQ-Z2)									
Method	Acc (%)	F1	Full App Latency (ms)	FPGA Kernel Latency (µs)	LUT	DSP	FF	BRAM	Speedup vs 810 ms
QKS-v2 S0	93.64	0.9621	18.5	1006	13,322	147	12,733	15	43.7×
QKS-v2 S10	93.79	0.9631	18.8	973	13,344	147	12,735	15	43.0×
QKS-v2 S20	94.11	0.9651	18.6	948	13,349	148	12,779	15	43.7×
QKS-v2 S30	93.84	0.9634	18.7	910	13,347	148	12,779	15	43.3×
ViT4Mal 1e2h (fastest)	92.41	—	810	—	16,545	69	16,911	92	1× (ref)
ViT4Mal 1e4h	92.47	—	870	—	16,029	69	16,358	84	0.9×
ViT4Mal 2e2h	93.86	—	1450	—	16,219	69	16,393	92	0.6×
ViT4Mal 2e4h (best acc)	93.89	—	1480	—	16,079	69	16,356	84	0.5×